B.TECH FIRST YEAR – ENGINEERING CHEMISTRY

Subject Code: CH101 | Credits: 3

UNIT II

# Modern Engineering Materials

## A Complete Study Module

From fundamentals to advanced applications — zero-level explanation with diagrams, solved numericals, and exam-oriented Q&A

⚡ Semiconductors | 🧲 Superconductors | 🔋 Supercapacitors | 🔬 Nanomaterials

⚽ Fullerenes | 🧵 Carbon Nanotubes | 🔷 Graphene

Prepared for B.Tech Students | Engineering Chemistry | Unit II

# Table of Contents

## PART F – QUESTIONS, NUMERICALS & REVISION

SECTION 1

# Introduction to Modern Engineering Materials

Why do we need advanced materials? | Historical context | What changed everything?

## 1.1 What Are Engineering Materials?

In engineering, **materials** are the physical substances from which devices, machines, structures, and components are made. From the Stone Age to Silicon Valley, human civilization has always been defined by the materials it uses.

💡 **THINK ABOUT IT**

Why does your smartphone work? Why can an MRI machine detect tumors without surgery? Why can a tiny chip control a rocket? The answer is always — **materials science**. The right material, engineered at the right scale, makes the impossible possible.

## 1.2 The Evolution of Engineering Materials

Engineering materials have evolved through several major eras:

| Era | Key Material | Defining Technology |
| --- | --- | --- |
| Stone Age (~3M BCE) | Stone, bone | Tools for hunting/farming |
| Bronze Age (~3300 BCE) | Bronze (Cu+Sn alloy) | Weapons, ornaments |
| Iron Age (~1200 BCE) | Iron, steel | Stronger tools & structures |
| Industrial Age (~1760 CE) | Steel, cast iron | Railways, bridges, engines |
| Silicon Age (~1950 CE) | Silicon semiconductors | Computers, electronics |
| Nano Age (~2000 CE) | Nanomaterials, superconductors | AI, quantum computing, energy |

## 1.3 Why Modern Engineering Materials?

Modern engineering demands materials with extraordinary properties that classical materials cannot provide:

### Ultra-fast Electronics

Semiconductors that switch billions of times per second — making your CPU possible.

### Zero Energy Loss

Superconductors that conduct electricity with zero resistance — enabling MRI, maglev trains.

### Instant Energy Storage

Supercapacitors that charge and discharge thousands of times faster than batteries.

### Atomic-Scale Engineering

Nanomaterials engineered atom by atom — with properties impossible in bulk form.

# SECTION 2

# Overview of Advanced Materials

Classification | Properties comparison | Big picture view

## 2.1 The Four Pillars of Unit II

This unit covers four categories of advanced engineering materials. Think of them as four different answers to the question: *"How can we make materials do more?"*

🗂️ **CLASSIFICATION AT A GLANCE**

**1. Semiconductors** — Materials whose conductivity lies between conductors and insulators, controlled by temperature, light, or impurities.

**2. Superconductors** — Materials that lose ALL electrical resistance below a critical temperature (Tc).

**3. Supercapacitors** — Energy storage devices that bridge the gap between capacitors and batteries.

**4. Nanomaterials** — Materials with at least one dimension in the range 1–100 nanometers, exhibiting unique quantum-scale properties.

## 2.2 Electrical Conductivity Comparison

**CONDUCTIVITY SPECTRUM: INSULATORS → SEMICONDUCTORS → CONDUCTORS**


*Fig 2.1: Electrical conductivity spectrum of materials (S/m = Siemens per metre)*

## 2.3 Quick Comparison Table

| Property | Semiconductor | Superconductor | Supercapacitor | Nanomaterial |
|---|---|---|---|---|
| Scale | Atomic (bulk) | Bulk material | Electrode device | 1–100 nm |
| Key feature | Tunable conductivity | Zero resistance | High capacitance | Quantum properties |
| Temperature | Room temp+ | Very low (mostly) | Room temp | Any |
| Examples | Si, Ge, GaAs | Hg, Nb, YBCO | Carbon-based | CNT, Graphene, C60 |
| Key application | Chips, solar cells | MRI, Maglev | EV, Power backup | Drug delivery, sensors |

SECTION 3

# Semiconductors – Basic Concepts & Band Theory

What is a semiconductor? | Energy band theory | Why silicon rules the world

## 3.1 Starting from Zero – What is Conductivity?

Before understanding semiconductors, we need to understand **why some materials conduct electricity and others don't**.

### Step 1: The Basic Idea — Electrons and Energy

Every atom has electrons orbiting its nucleus. For a material to conduct electricity, **electrons must be free to move**. In metals like copper, the outermost electrons are already free — this is why metals conduct well.

But in materials like glass or rubber, ALL electrons are tightly bound — no free electrons means no conductivity.

### Step 2: The Energy Band Model

When millions of atoms come together in a solid, their individual energy levels merge into broad **energy bands**:

- **Valence Band (VB):** The highest energy band that is completely filled with electrons at absolute zero (0 K). Electrons here are bound to atoms.
- **Conduction Band (CB):** The next higher energy band, which is empty (or partly filled). Electrons here are FREE to conduct electricity.
- **Forbidden Gap (Band Gap, Eg):** The energy gap between the top of the valence band and the bottom of the conduction band. Electrons CANNOT exist here.

📖 DEFINITION

A **Semiconductor** is a material whose electrical conductivity lies between that of a conductor and an insulator. The band gap (Eg) of a semiconductor is typically between 0.1 eV and 3.0 eV, and its conductivity increases with temperature.

## 3.2 Energy Band Diagrams (The Most Important Diagram!)

**ENERGY BAND STRUCTURE: CONDUCTOR VS SEMICONDUCTOR VS INSULATOR**

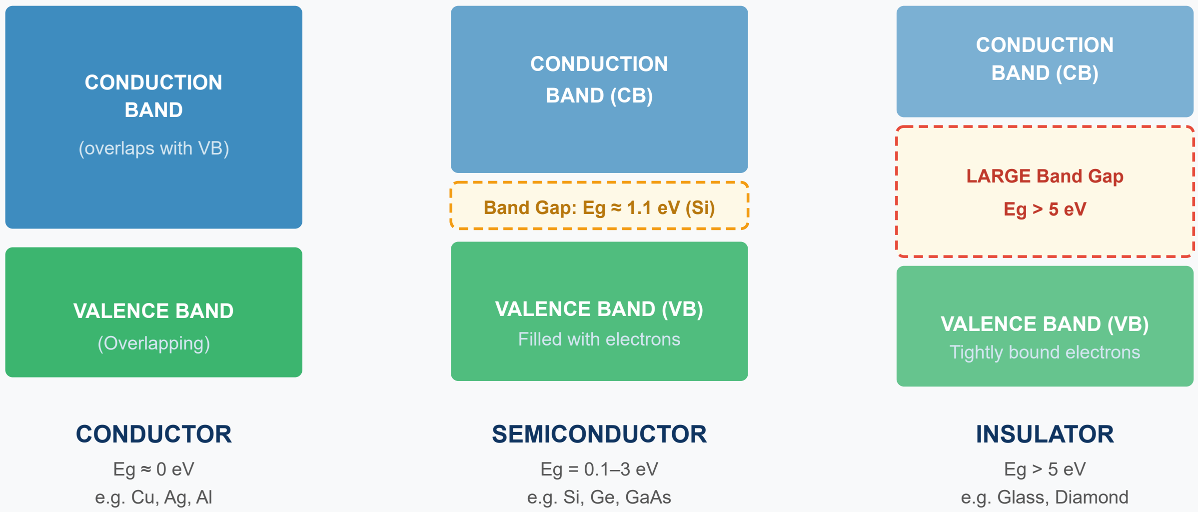



*Fig 3.1: Energy band diagrams for conductor, semiconductor, and insulator. The forbidden gap (Eg) determines electrical behavior.*

## 3.3 Band Gap Values of Common Semiconductors

| Material | Symbol | Band Gap (eV) | Type | Key Application |
|---|---|---|---|---|
| Silicon | Si | 1.12 | Indirect | Microchips, solar cells |
| Germanium | Ge | 0.67 | Indirect | IR detectors, transistors |
| Gallium Arsenide | GaAs | 1.42 | Direct | LEDs, laser diodes |
| Gallium Nitride | GaN | 3.36 | Direct | Blue LEDs, power devices |
| Silicon Carbide | SiC | 2.36–3.23 | Indirect | High-temp power electronics |
| Indium Phosphide | InP | 1.35 | Direct | Optical fiber comm. |

⚡ KEY FORMULAS – SEMICONDUCTOR CONDUCTIVITY

$$\sigma = n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h$$

$$\sigma = \sigma_0 \cdot \exp(-Eg / 2kT)$$

$$n \cdot p = ni^2 \text{ (Mass Action Law)}$$

Where: $\sigma$ = conductivity (S/m) | n = electron concentration ($m^{-3}$) | p = hole concentration ($m^{-3}$)

e = electron charge = $1.6 \times 10^{-19}$ C | $\mu_e$ = electron mobility | $\mu_h$ = hole mobility

Eg = band gap energy (eV) | k = Boltzmann constant = $8.617 \times 10^{-5}$ eV/K | T = temperature (K)

ni = intrinsic carrier concentration

## 3.4 Effect of Temperature on Semiconductor Conductivity

This is fundamentally different from metals:

⚠️ **CRITICAL DIFFERENCE**

**Metals:** Conductivity DECREASES with increasing temperature (more atomic vibrations disrupt electron flow).

**Semiconductors:** Conductivity INCREASES with increasing temperature. At higher temperatures, more electrons gain enough energy to jump the band gap — creating more free carriers.

SECTION 4

# Types of Semiconductors

Intrinsic | Extrinsic (n-type & p-type) | Doping concept | p-n Junction

## 4.1 Intrinsic Semiconductors

📖 DEFINITION

An **Intrinsic Semiconductor** is a pure semiconductor material with no intentional impurities added. Its electrical properties depend entirely on its own atomic structure and temperature.

**Step-by-Step: How Conduction Occurs in Intrinsic Si**

**Step 1:** Silicon has 4 valence electrons. In a pure Si crystal, each Si atom forms 4 covalent bonds with neighbors.

**Step 2:** At 0 K (absolute zero), ALL electrons are in covalent bonds — NO free electrons — Si acts like an insulator.

**Step 3:** At room temperature (~300 K), some electrons gain thermal energy and BREAK their covalent bonds.

**Step 4:** A freed electron goes to the Conduction Band. The empty spot left behind is called a **HOLE** — it acts like a positive charge carrier.

**Step 5:** Both electrons (in CB) and holes (in VB) contribute to current flow. This is called **bipolar conduction**.

**Key property:** In intrinsic semiconductors, the number of electrons equals the number of holes: **$n = p = n_i$** (intrinsic carrier concentration).

## 4.2 Extrinsic Semiconductors (Doped Semiconductors)

The conductivity of intrinsic semiconductors is too low for practical use. By adding tiny amounts of specific impurities (called **dopants**), we can dramatically increase conductivity. This process is called **DOPING**.

> 🔬 **WHAT IS DOPING?**
>
> Doping means intentionally adding impurity atoms (in parts per million or billion) to a pure semiconductor crystal to modify its electrical properties. Even 1 dopant atom per 10 million Si atoms can increase conductivity by a million times!

### 4.2.1 N-type Semiconductor (Donor Doping)

**How N-type is Created**

**Dopant used:** Group V elements (Pentavalent) — Phosphorus (P), Arsenic (As), Antimony (Sb)

**Process:** A Group V atom has 5 valence electrons. When it replaces a Si atom (which has 4 bonds), 4 electrons form bonds with neighboring Si atoms. The 5th electron is loosely held and becomes a FREE electron with very little energy input.

**Result:** $n >> p$ — Electrons are MAJORITY carriers, holes are minority carriers. Called "n-type" because negative carriers dominate.

**Donor atoms** — they DONATE electrons to the conduction band.

### 4.2.2 P-type Semiconductor (Acceptor Doping)

**How P-type is Created**

**Dopant used:** Group III elements (Trivalent) — Boron (B), Gallium (Ga), Indium (In)

**Process:** A Group III atom has 3 valence electrons. When it replaces a Si atom, it forms only 3 covalent bonds, leaving one bond incomplete — this creates a HOLE.

**Result:** $p >> n$ — Holes are MAJORITY carriers, electrons are minority carriers. Called "p-type" because positive carriers dominate.

**Acceptor atoms** — they ACCEPT electrons from the valence band (equivalently, they create holes).

**INTRINSIC VS N-TYPE VS P-TYPE SEMICONDUCTOR — VISUAL COMPARISON**

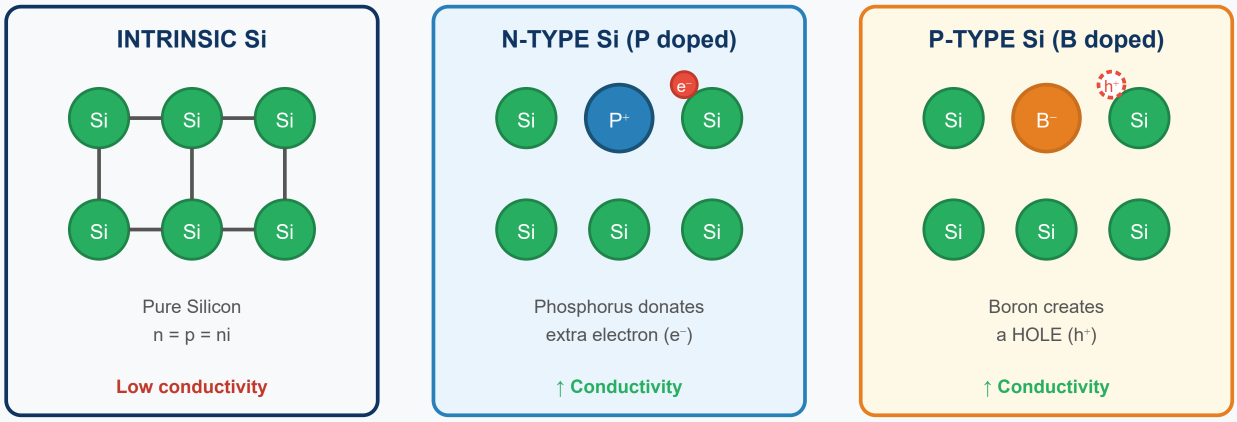



*Fig 4.1: Crystal structure comparison — Intrinsic Si, N-type (P-doped), P-type (B-doped). Red dot = free electron, white circle = hole.*

## 4.3 p-n Junction

When a p-type and n-type semiconductor are joined together, a **p-n junction** is formed. This is the fundamental building block of all semiconductor devices.

P-N JUNCTION FORMATION & DEPLETION REGION

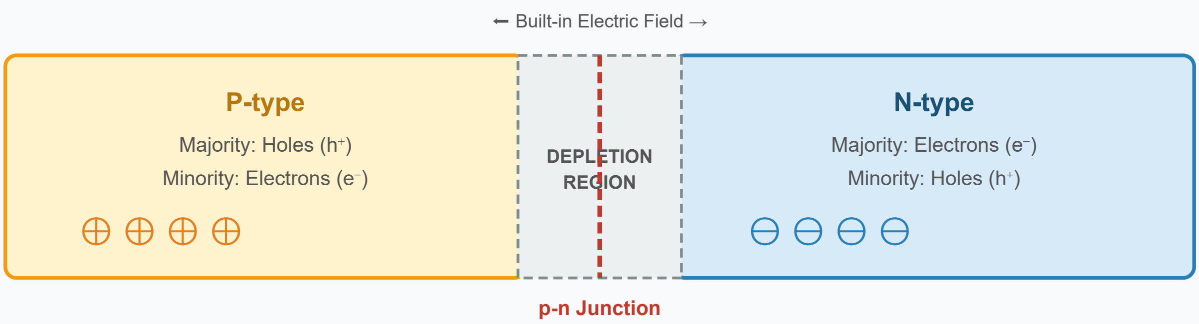



*Fig 4.2: p-n Junction showing p-type, n-type regions and the depletion zone (space charge region) at the junction interface.*

## 4.4 Comparison: Intrinsic vs Extrinsic Semiconductors

| Property | Intrinsic | N-type (Extrinsic) | P-type (Extrinsic) |
|---|---|---|---|
| Purity | Chemically pure | Doped with Group V | Doped with Group III |
| Majority carriers | $e^- = h^+ = n_i$ | Electrons ($e^-$) | Holes ($h^+$) |
| Minority carriers | Same | Holes | Electrons |

| | | | |
|---|---|---|---|
| Dopant examples | None | P, As, Sb | B, Ga, In |
| Conductivity | Low | High | High |
| Fermi level | Mid-gap | Near CB | Near VB |

SECTION 5

# Applications of Semiconductors

From transistors to solar cells — semiconductors are everywhere

## 5.1 Major Applications

### 🖥️ Integrated Circuits (ICs)

Billions of transistors on a chip smaller than a fingernail. Enables processors, memory, and computing.

### 🌞 Solar Cells

p-n junction converts photons to electricity via photovoltaic effect. Si-based solar panels are ~22% efficient.

### 💡 Light Emitting Diodes (LEDs)

GaAs, GaN junctions emit light when forward biased. Used in displays, lighting, indicators.

### 📡 Photodetectors

Detect light by generating electron-hole pairs. Used in cameras, fiber optics, medical imaging.

### 🔉 Transistors

The fundamental amplifying/switching device. MOSFET is the most common — forms the basis of all digital electronics.

### 🌡️ Thermistors & Sensors

Temperature-sensitive resistors for temperature measurement, medical diagnostics, and automotive sensors.

**🎯 Exam Key Points – Semiconductors**

- Semiconductor conductivity INCREASES with temperature (opposite to metals)
- Band gap of Si = 1.12 eV, Ge = 0.67 eV

- N-type: Group V dopants (P, As, Sb) → excess electrons
- P-type: Group III dopants (B, Ga, In) → holes created
- At p-n junction: depletion region forms due to recombination of carriers
- Hall effect: used to determine type (n or p) and carrier concentration
- Conductivity: $\sigma = n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h$

SECTION 6

# Superconductors – Concept & Properties

Zero resistance | Meissner effect | Critical temperature | Cooper pairs

## 6.1 The Discovery That Changed Physics

In 1911, Dutch physicist **Heike Kamerlingh Onnes** cooled mercury (Hg) to 4.2 K (−268.95°C) and found something astonishing: its electrical resistance suddenly dropped to EXACTLY ZERO. This phenomenon became known as **superconductivity**.

💡 **EVERYDAY ANALOGY**

Imagine a highway with zero traffic resistance — cars (electrons) move forever without any braking, no energy lost, no heat generated. That's what happens in a superconductor. Once a current starts, it flows *forever* without any driving voltage!

📖 **DEFINITION**

A **Superconductor** is a material that, when cooled below a characteristic temperature called the *Critical Temperature (Tc)*, exhibits zero electrical resistance and expels all magnetic flux from its interior (Meissner Effect).

## 6.2 Critical Temperature (Tc)

The critical temperature is the temperature below which a material becomes superconducting. Above Tc, the material behaves like a normal conductor (with finite resistance).

| Material | Symbol | Critical Temp (Tc) | Type |
|---|---|---|---|
| Mercury | Hg | 4.15 K | Type I |
| Lead | Pb | 7.19 K | Type I |
| Niobium | Nb | 9.25 K | Type II |

| | | | |
|---|---|---|---|
| Niobium-Titanium | NbTi | 10 K | Type II |
| YBCO (Yttrium Barium Copper Oxide) | $YBa_2Cu_3O_7$ | 92 K | Type II (HTS) |
| BSCCO (Bismuth Strontium Calcium Copper Oxide) | $Bi_2Sr_2Ca_2Cu_3O_{10}$ | 110 K | Type II (HTS) |
| Mercury Cuprate | $HgBa_2Ca_2Cu_3O_8$ | 138 K (record) | Type II (HTS) |

## 6.3 Resistance vs Temperature Graph

**RESISTANCE VS TEMPERATURE — NORMAL CONDUCTOR VS SUPERCONDUCTOR**

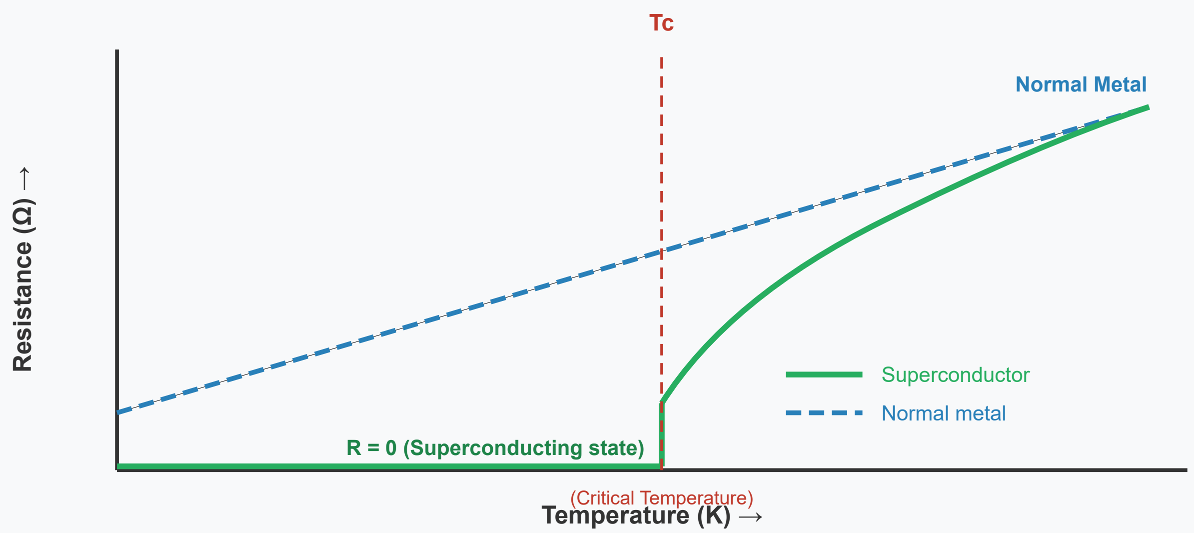



*Fig 6.1: Resistance vs Temperature graph. Below Tc, resistance drops abruptly to ZERO in superconductors. Normal metals show gradual decrease but never reach zero.*

## 6.4 The Meissner Effect

**What is the Meissner Effect?**

When a material transitions to its superconducting state (below Tc), it **expels ALL magnetic flux** from its interior. This means a superconductor is a **perfect diamagnet** — it actively repels magnetic fields.

**Practical demonstration:** A magnet placed on a superconductor LEVITATES above it. The superconductor induces surface currents that create a repelling magnetic field exactly equal to the applied field.

**MEISSNER EFFECT — MAGNETIC FLUX EXPULSION IN SUPERCONDUCTOR**

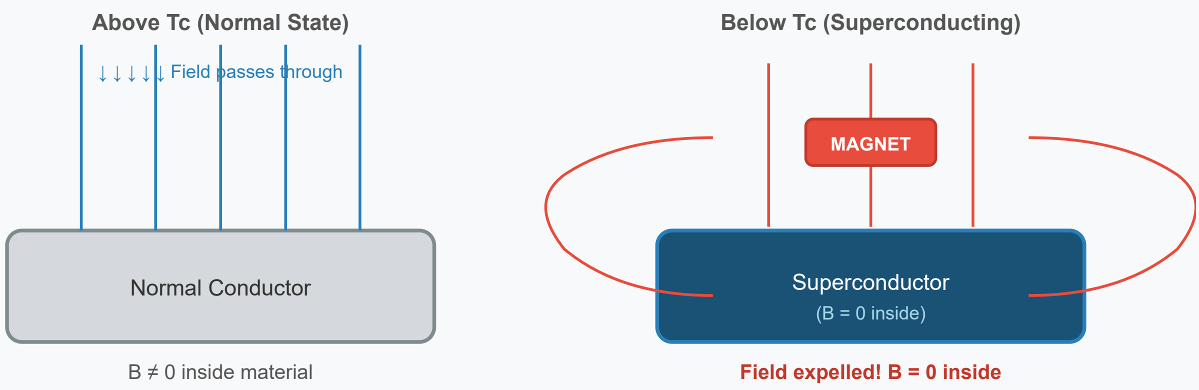



*Fig 6.2: Meissner Effect — Above Tc, magnetic field penetrates normally. Below Tc, the superconductor expels ALL magnetic flux (Meissner Effect), causing magnetic levitation.*

## 6.5 BCS Theory (Cooper Pairs) — Microscopic Explanation

In 1957, Bardeen, Cooper, and Schrieffer proposed the **BCS Theory** to explain superconductivity at the atomic level.

### How BCS Theory Works

**Step 1:** When an electron moves through a crystal lattice, it attracts nearby positive ions toward it, creating a region of slightly higher positive charge density.

**Step 2:** This region of positive charge attracts a second electron, creating a weak attractive force between the two electrons.

**Step 3:** The two electrons form a bound pair called a **Cooper pair**. Cooper pairs have zero net spin and are bosons.

**Step 4:** Cooper pairs move through the lattice as a coherent quantum state — they are not scattered by impurities or lattice vibrations. Hence, ZERO resistance!

⚡ KEY FORMULAS – SUPERCONDUCTORS

$$H_c(T) = H_c(0)[1 - (T/T_c)^2]$$

$$\lambda(T) = \lambda(0) / [1 - (T/T_c)^4]^{(1/2)}$$

London Penetration Depth: λ(T) = λ(0) / [1 − (T/Tc)⁴]^(1/2)

Meissner Effect: $B = 0$ (inside superconductor)

Where: $H_c$ = Critical magnetic field | $H_c(0)$ = Critical field at 0 K

$T$ = temperature | $T_c$ = critical temperature | $\lambda$ = London penetration depth

$B$ = Magnetic flux density

## 6.6 Properties of Superconductors

| Property | Description |
|---|---|
| **Zero Electrical Resistance** | $R = 0$ below $T_c$. Currents flow indefinitely without energy loss. |
| **Meissner Effect** | Complete expulsion of magnetic field from interior. Perfect diamagnetism. |
| **Critical Temperature (Tc)** | Temperature below which superconductivity occurs. Ranges from 4 K to 138 K. |
| **Critical Magnetic Field (Hc)** | Above this field strength, superconductivity is destroyed. |
| **Critical Current Density (Jc)** | Maximum current the superconductor can carry while remaining superconducting. |
| **Isotope Effect** | $T_c \propto M^{(-1/2)}$ where M is isotopic mass. Evidence for phonon-mediated pairing. |
| **Josephson Effect** | Quantum tunneling of Cooper pairs across a thin insulating barrier. |
| **Flux Quantization** | Magnetic flux through a superconducting loop is quantized in units of $\Phi_0 = h/2e$. |

SECTION 7

# Types of Superconductors

Type I vs Type II | LTS vs HTS | Vortex state

## 7.1 Type I Superconductors

Type I superconductors are simple elemental metals (like Hg, Pb, Sn) that exhibit a sharp, complete transition from superconducting to normal state at a single critical magnetic field (Hc). Below Hc, they are perfect diamagnets. Above Hc, superconductivity is completely destroyed.

**TYPE I PROPERTIES**

• Have a single critical field (Hc) — typically low values (10–200 mT)
• Complete Meissner effect below Hc
• Abrupt transition to normal state above Hc
• Examples: Mercury (Hg, Tc=4.15K), Lead (Pb, Tc=7.19K), Tin (Sn, Tc=3.72K), Aluminum (Al, Tc=1.17K)
• Described well by BCS theory

## 7.2 Type II Superconductors

Type II superconductors have TWO critical fields: Hc1 (lower) and Hc2 (upper). Between Hc1 and Hc2, magnetic flux partially penetrates in quantized vortices — this is the "vortex state" or "mixed state."

**TYPE II PROPERTIES**

• Two critical fields: Hc1 and Hc2 (Hc2 can be extremely high — up to 100 T in YBCO!)
• Vortex/mixed state between Hc1 and Hc2 (partial flux penetration)
• Complete superconductivity below Hc1
• Normal state above Hc2
• Examples: NbTi, YBCO, BSCCO (used in practical applications)
• High-Temperature Superconductors (HTS) are all Type II

## 7.3 Comparison: Type I vs Type II

| Property | Type I | Type II |
|---|---|---|
| Critical field | Single Hc | Two fields: Hc1 and Hc2 |
| Meissner effect | Complete below Hc | Complete below Hc1; partial Hc1–Hc2 |
| Vortex state | Not present | Present between Hc1 and Hc2 |
| Tc range | Usually < 10 K | Up to 138 K (HTS) |
| Materials | Pure metals (Hg, Pb, Sn) | Alloys and ceramics (NbTi, YBCO) |
| Practical use | Limited | Widely used in technology |

## 7.4 LTS vs HTS (Temperature Classification)

| Property | Low Temp Superconductors (LTS) | High Temp Superconductors (HTS) |
|---|---|---|
| Tc range | < 30 K | 30 K – 138 K |
| Cooling medium | Liquid Helium (4.2 K) | Liquid Nitrogen (77 K) — cheaper! |
| Theory | BCS theory | Not fully explained by BCS |
| Examples | Nb, NbTi, $Nb_3Sn$ | YBCO, BSCCO, TBCCO |
| Discovery | 1911 onwards | 1986 (Bednorz and Müller) |

SECTION 8

# Applications of Superconductors

MRI | Maglev trains | Power cables | Quantum computing

## 8.1 Key Applications

| Application | How Superconductivity Helps | Example |
|---|---|---|
| **MRI Machines** | Superconducting coils create powerful, stable magnetic fields without energy loss | NbTi coils at 4 K in hospital MRI scanners |
| **Maglev Trains** | Meissner effect provides contactless levitation — no friction, ultra-high speed | Japan's SCMaglev (603 km/h world record) |
| **Particle Accelerators** | Superconducting magnets guide charged particles in circular paths | CERN's Large Hadron Collider (LHC) |
| **Power Transmission** | Zero resistance means zero power loss in transmission cables | HTS power cables in cities |
| **SQUIDs** | Superconducting Quantum Interference Devices detect ultra-tiny magnetic fields | Brain mapping (MEG), geological surveys |
| **Electric Motors** | Superconducting motors are 50% lighter and more efficient than conventional ones | Ship propulsion systems |
| **Quantum Computing** | Josephson junction-based qubits maintain quantum coherence longer | IBM, Google quantum computers |
| **Energy Storage** | SMES (Superconducting Magnetic Energy Storage) stores energy in magnetic field | Grid stabilization |

🎯 **Exam Key Points – Superconductors**

- Discovered by Kamerlingh Onnes in 1911 in mercury at 4.15 K
- Two signatures: (1) Zero resistance and (2) Meissner effect

- BCS theory: Cooper pairs are the current carriers (electron pairs bound by phonon exchange)
- Type I: single Hc, pure metals; Type II: two critical fields, used practically
- HTS (High Tc): YBCO at 92 K, can use liquid $N_2$ cooling (economical)
- Critical parameters: Tc, Hc (or Hc1, Hc2), Jc — called the "critical triangle"

SECTION 9

# Supercapacitors – Introduction & Concept

Beyond ordinary capacitors | Energy storage between batteries and capacitors

## 9.1 Why Do We Need Supercapacitors?

To understand supercapacitors, we first need to understand the energy storage landscape. Engineers need devices that can store energy AND deliver it quickly.

⚡ **THE ENERGY STORAGE PROBLEM**

**Batteries:** High energy density (store lots of energy) but SLOW to charge/discharge. Charge cycles are limited (~500–2000 cycles for Li-ion).

**Capacitors:** Extremely fast charge/discharge but VERY low energy density. Can be cycled millions of times.

**Supercapacitors:** Bridge the gap — much higher energy than capacitors, much faster than batteries. Over 1 million charge cycles!

📖 DEFINITION

A **Supercapacitor** (also called *Electrochemical Double Layer Capacitor / EDLC / Ultracapacitor*) is an energy storage device that stores charge electrostatically at the electrode-electrolyte interface (through double layer formation and/or fast faradaic reactions), achieving capacitances of 1–10,000 Farads — thousands of times more than conventional capacitors.

## 9.2 Ragone Plot — The Big Picture Comparison

**RAGONE PLOT: ENERGY DENSITY VS POWER DENSITY**

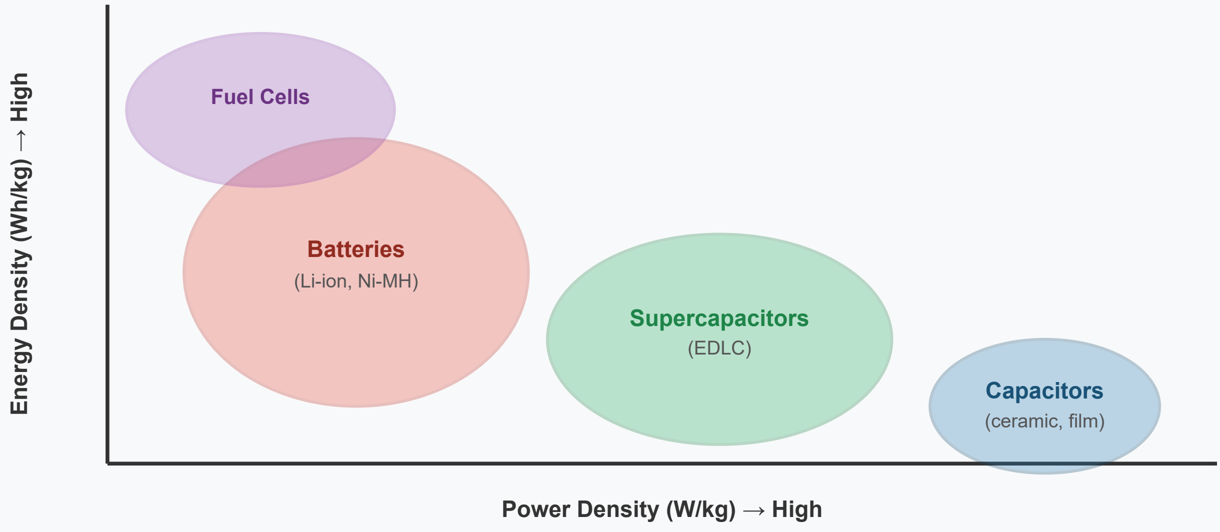



*Fig 9.1: Ragone Plot showing energy vs. power density. Supercapacitors occupy the "sweet spot" between batteries (high energy) and capacitors (high power).*

## 9.3 Basic Structure of a Supercapacitor

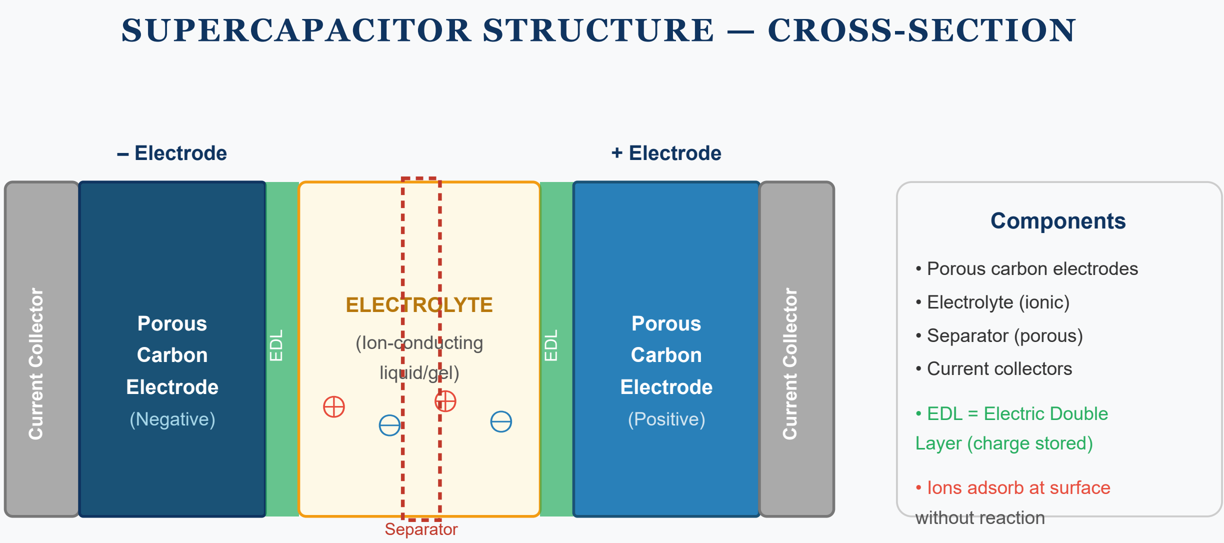



*Fig 9.2: Cross-section of a supercapacitor. Charge is stored in the Electric Double Layer (EDL) formed at the electrode-electrolyte interface. No chemical reactions occur during charging.*

SECTIONS 10 & 11

# Classification & Working Mechanism of Supercapacitors

EDLC | Pseudocapacitors | Hybrid | How energy is stored

## 10.1 Classification of Supercapacitors

Supercapacitors are classified based on their charge storage mechanism into three main types:

### 10.1.1 Electric Double Layer Capacitors (EDLCs)

**What are EDLCs?**

**Mechanism:** Non-Faradaic — stores charge purely electrostatically (no chemical reactions). Ions from the electrolyte adsorb on the surface of highly porous carbon electrodes.

**Electrode materials:** Activated carbon, carbon aerogel, carbon nanotubes, graphene

**Advantages:** Very long cycle life (1 million+ cycles), fast charge/discharge, no degradation

**Disadvantage:** Lower energy density than pseudocapacitors

### 10.1.2 Pseudocapacitors

**What are Pseudocapacitors?**

**Mechanism:** Faradaic — stores charge through rapid, reversible redox reactions at the electrode surface (like a battery, but at the surface only — not in bulk).

**Electrode materials:** $RuO_2$ (ruthenium oxide), $MnO_2$, conducting polymers (PANI, PPy)

**Advantages:** Higher energy density than EDLC

**Disadvantage:** Shorter cycle life than EDLC, slower kinetics

### 10.1.3 Hybrid Supercapacitors

### What are Hybrid Supercapacitors?

**Mechanism:** Combines EDLC electrode (carbon) with a battery-type or pseudocapacitor electrode. Gets the best of both: high energy AND high power.

**Types:** Composite hybrid, Asymmetric hybrid, Battery-type hybrid

**Example:** Lithium-ion capacitor — EDLC cathode + Li-intercalation anode

| Type | Mechanism | Materials | Energy | Power | Cycle Life |
|---|---|---|---|---|---|
| EDLC | Non-Faradaic (EDL) | Activated Carbon | Low | Very High | ~1M cycles |
| Pseudocapacitor | Faradaic (surface redox) | $RuO_2$, $MnO_2$ | Medium | High | ~100K cycles |
| Hybrid | Both | Carbon + metal oxide | High | High | ~500K cycles |

## 11.1 Working Mechanism of EDLCs (Step-by-Step)

### Step 1 — Charging Phase

When voltage is applied, positive electrode attracts negative ions (anions) from electrolyte; negative electrode attracts positive ions (cations). These ions pack densely at the electrode surface, forming the Electric Double Layer.

### Step 2 — Charge Storage

The charge is stored as electrostatic potential energy — similar to a parallel plate capacitor but with nanometer-scale separation between electrode surface and ion layer. This tiny gap (0.5–1 nm) with high surface area (1000–3000 $m^2/g$ for activated carbon) gives enormous capacitance.

### Step 3 — Discharge Phase

When connected to a load, ions desorb from the electrode surface and move back into electrolyte. Electrons flow through external circuit. No chemical reactions occur — fully reversible process.

⚡ KEY FORMULAS – SUPERCAPACITORS

$$C = \varepsilon_0 \cdot \varepsilon_r \cdot A / d$$

$$E = \frac{1}{2} \cdot C \cdot V^2$$

$$P = V^2 / (4 \cdot ESR)$$

Where: C = Capacitance (Farads) | $\varepsilon_0$ = permittivity of free space | $\varepsilon_r$ = relative permittivity

A = Electrode surface area ($m^2$) | d = double layer thickness (~0.5–1 nm)

E = Energy stored (Joules) | V = Voltage (Volts) | P = Power (Watts)

ESR = Equivalent Series Resistance

## 11.2 Applications of Supercapacitors

| Application | Why Supercapacitor is Ideal |
|---|---|
| Electric Vehicles (EVs) | Regenerative braking — captures burst energy during braking, releases during acceleration |
| Start-Stop Engines | Provides instant high power for engine restarts — extends battery life |
| Wind/Solar Energy | Smooths fluctuating power output from renewable sources |
| UPS (Uninterruptible Power Supply) | Instant power backup for critical systems (hospitals, data centers) |
| Consumer Electronics | Extends battery life in phones, wearables by handling power spikes |
| Military & Aerospace | Reliable, long-lasting power for weapons systems, satellites |
| Smart Grids | Grid stabilization and frequency regulation |

SECTION 12

# Nanomaterials – Introduction & Classification

The world at 1–100 nm | Quantum effects | Why small is different

## 12.1 What Does "Nano" Mean?

The prefix "nano" comes from the Greek word *nanos* meaning "dwarf." In science, nano = $10^{-9}$ meters. To put this in perspective:

**SIZE COMPARISON — FROM EVERYDAY OBJECTS TO NANOSCALE**

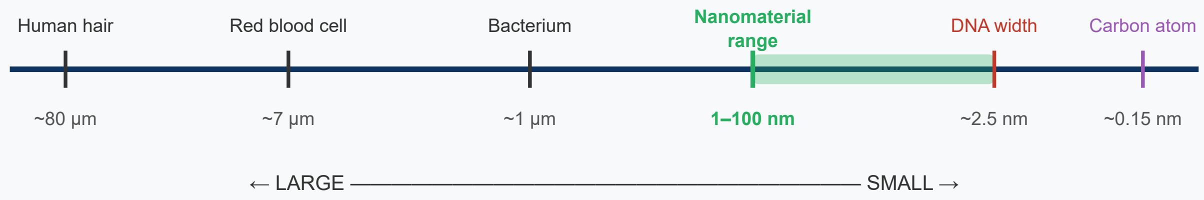



*Fig 12.1: Size scale comparison. Nanomaterials (1–100 nm) are between the size of individual molecules and bacteria.*

📖 DEFINITION

**Nanomaterials** are materials having at least one dimension in the nanometer range (1–100 nm). At this scale, quantum mechanical effects dominate, and materials exhibit dramatically different physical, chemical, optical, electrical, and magnetic properties compared to their bulk counterparts.

## 12.2 Why Are Nanomaterials Special? (The Quantum Size Effect)

**Why Small Size Changes Everything**

**1. Surface Area Dominates:** A 10 nm particle has ~40% of its atoms on the surface. A 1 µm particle has only ~0.04% on the surface. More surface atoms → dramatically different reactivity.

**2. Quantum Confinement:** When electrons are confined in nanoscale dimensions, their energy levels become quantized (discrete) — like electrons in atoms. This changes optical and electronic properties fundamentally.

**3. Classical Physics Fails:** At nanoscale, quantum mechanics governs behavior. Properties like color, melting point, hardness, conductivity become size-dependent!

**Example:** Gold nanoparticles are RED (not golden) due to surface plasmon resonance. The melting point of gold drops from 1064°C in bulk to ~300°C at 2 nm size!

## 12.3 Classification of Nanomaterials

Nanomaterials are classified by the number of dimensions that are NOT in the nanoscale:

| Dimensionality | Name | Description | Examples |
|---|---|---|---|
| **0D** (Zero-dimensional) | Nanoparticles / Quantum Dots | All 3 dimensions at nanoscale. Essentially a tiny sphere/cluster. | Gold nanoparticles, $C_{60}$ Fullerene, Quantum dots |
| **1D** (One-dimensional) | Nanowires / Nanotubes / Nanorods | One dimension extends beyond nanoscale (length). Width/diameter still nano. | Carbon Nanotubes (CNT), ZnO nanowires, Si nanorods |
| **2D** (Two-dimensional) | Nanosheets / Thin films | Two dimensions extend beyond nanoscale. Thickness is nano. | Graphene, $MoS_2$, h-BN, Thin films |
| **3D** (Three-dimensional) | Bulk nanostructured materials | All dimensions > 100 nm but grain size at nanoscale. Nano-grained. | Nanocrystalline metals, nano-composites |

## 12.4 Nanomaterial Properties — Bulk vs Nano Comparison

| Property | Bulk Material | Nanomaterial |
|---|---|---|

| | | |
|---|---|---|
| Melting point | Fixed (e.g., Au: 1064°C) | Strongly size-dependent (lower at small sizes) |
| Color | Fixed (Au: gold) | Changes with size (Au nano: red/blue) |
| Hardness | Fixed | Can be much harder (nano-steel, etc.) |
| Electrical conductivity | Fixed | Can change dramatically (CNT: metallic or semiconducting) |
| Chemical reactivity | Surface area-limited | Extremely high (huge surface-to-volume ratio) |
| Optical properties | Bulk-determined | Quantum confinement — tunable |
| Magnetic properties | Ferromagnetic (Fe) | Superparamagnetic at < ~20 nm |

SECTION 13

# Fullerenes – The Soccer Ball Molecule

$C_{60}$ | Buckminsterfullerene | Structure | Properties | Applications

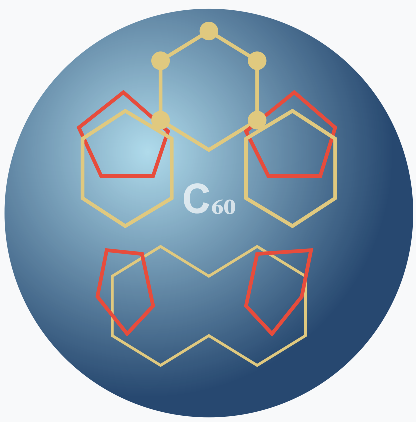
## 13.1 Discovery

Fullerenes were discovered in 1985 by **Kroto, Curl, and Smalley** (Nobel Prize in Chemistry, 1996). The most famous fullerene, $C_{60}$, was named **Buckminsterfullerene** after architect Richard Buckminster Fuller whose geodesic domes inspired the structure.

> 📖 DEFINITION
>
> **Fullerenes** are allotropes of carbon consisting of carbon atoms arranged in closed cage structures of pentagons and hexagons. The most stable and common is $C_{60}$, which looks exactly like a soccer ball.

## 13.2 Structure of $C_{60}$

**FULLERENE $C_{60}$ (BUCKMINSTERFULLERENE) STRUCTURE**




**$C_{60}$ Key Facts**

- 20 Hexagons (6-membered)
- 12 Pentagons (5-membered)

Total: 60 Carbon atoms
Diameter: ~0.7 nm
Hybridization: $sp^2$ + $sp^3$
Structure: Truncated icosahedron
Other: $C_{70}$, $C_{84}$, $C_{240}$...

*Fig 13.1: Structure of $C_{60}$ Buckminsterfullerene — 20 hexagons (yellow) + 12 pentagons (red) = 60 C atoms in a hollow spherical cage, like a soccer ball.*

## 13.3 Properties of Fullerenes

| Property | Description |
|---|---|
| Appearance | Yellow-brown crystalline solid ($C_{60}$ in pure form) |
| Structure | Hollow cage of 60 C atoms; diameter ~0.7 nm |
| Hybridization | Intermediate between $sp^2$ and $sp^3$ (due to cage curvature) |
| Solubility | Soluble in organic solvents like toluene, $CS_2$ (purple solution) |
| Electrical | Semiconductor in pure form; superconductor when $K_3C_{60}$ (at 18 K) |
| Stability | Thermally stable up to ~600°C; chemically stable; can withstand high pressure |
| Encapsulation | Can trap atoms/molecules inside cage — forms "endohedral fullerenes" |
| Reactivity | Undergoes addition reactions — acts as electron acceptor |

## 13.4 Applications of Fullerenes

### 💊 Drug Delivery

Drug molecules can be encapsulated inside $C_{60}$ cage or attached to surface. Highly targeted cancer therapy.

### 🌞 Solar Cells

$C_{60}$ derivatives (like PCBM) are used as electron acceptors in organic photovoltaic cells.

### 🔋 Batteries

Fullerene-based electrode materials for Li-ion batteries — high capacity and stability.

### 🧴 Cosmetics

Anti-aging creams — $C_{60}$ is a powerful antioxidant that neutralizes free radicals.

SECTION 14

# Carbon Nanotubes (CNTs)

SWCNT vs MWCNT | The strongest material known | Electronic highways

## 14.1 Discovery & Introduction

Carbon Nanotubes were discovered by **Sumio Iijima** in 1991 (MWCNT) and in 1993 (SWCNT). They can be thought of as a graphene sheet rolled into a seamless cylinder.

📖 DEFINITION

A **Carbon Nanotube (CNT)** is a hollow cylindrical nanostructure made of carbon atoms arranged in hexagonal lattice. The diameter is typically 1–50 nm while the length can be several millimeters — giving them an extraordinary aspect ratio.

## 14.2 Types of CNTs

**SWCNT VS MWCNT — STRUCTURAL COMPARISON**

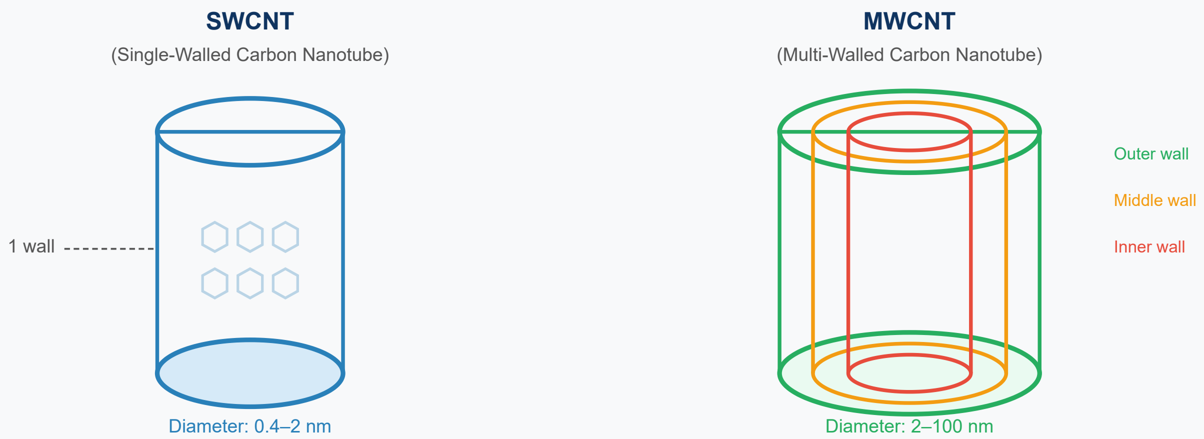



*Fig 14.1: SWCNT (left) — single hexagonal graphene wall cylinder. MWCNT (right) — multiple concentric graphene cylinders like nested test tubes.*

## 14.3 Properties of Carbon Nanotubes

| Property | SWCNT | MWCNT |
|---|---|---|
| Diameter | 0.4–2 nm | 2–100 nm |
| Tensile strength | ~63 GPa (100× steel!) | ~150 GPa |
| Young's modulus | ~1 TPa | ~0.1–0.9 TPa |
| Electrical conductivity | Metallic or semiconducting (chirality-dependent) | Mostly metallic |
| Thermal conductivity | ~3500 W/m·K (exceeds diamond!) | ~3000 W/m·K |
| Density | ~1.3 g/cm³ (very light) | ~2.1 g/cm³ |
| Band gap | 0 (metallic) or 0.4–2 eV (semiconducting) | Essentially 0 (metallic) |

🏆 **CNT WORLD RECORDS**

• **Strongest material per unit weight** — 100× stronger than steel at 1/6 the weight
• **Best thermal conductor** — better than diamond (at room temperature)
• **Ultimate electrical wire** — metallic CNTs carry current density 1000× greater than copper without melting

## 14.4 Chirality — What Makes a CNT Metallic or Semiconducting?

The electrical behavior of SWCNT depends on the angle (chirality) at which the graphene sheet is rolled:

🔬 **CHIRALITY RULE**

A SWCNT is described by chiral vector (n, m):
• If **n = m** → Armchair CNT → **METALLIC**
• If **n − m = 3k** (integer k) → **METALLIC**
• All others → **SEMICONDUCTING**
Roughly 1/3 of SWCNTs are metallic and 2/3 are semiconducting.

## 14.5 Applications of CNTs

| Application Area | How CNTs Help |
|---|---|
| Structural composites | Add to polymers/metals for ultra-strong lightweight materials (aerospace, sports equipment) |
| Electronics | CNT transistors, interconnects — potential successor to silicon-based chips |
| Energy storage | CNT electrodes in supercapacitors and batteries — high surface area |
| Medical | Drug delivery vehicles, cancer cell targeting, biosensors |
| Thermal management | Heat spreaders in electronics — high thermal conductivity |
| Sensors | Detect single molecules — gas sensors, chemical sensors, biosensors |
| Field emission | Sharp CNT tip → excellent field emitters for flat panel displays, electron guns |

SECTION 15

# Graphene – The Wonder Material

The thinnest material | The strongest material | The best conductor

## 15.1 Discovery

Graphene was isolated in 2004 by **Andre Geim and Konstantin Novoselov** at the University of Manchester using *Scotch tape* (mechanical exfoliation from graphite). They were awarded the **Nobel Prize in Physics in 2010**.

📖 DEFINITION

**Graphene** is a single atomic layer of carbon atoms arranged in a two-dimensional hexagonal (honeycomb) lattice. It is the thinnest material ever isolated — exactly ONE atom thick (~0.335 nm). It is the parent material for all other carbon allotropes: roll it → CNT, wrap it → fullerene, stack it → graphite.

## 15.2 Structure of Graphene

GRAPHENE HONEYCOMB LATTICE STRUCTURE

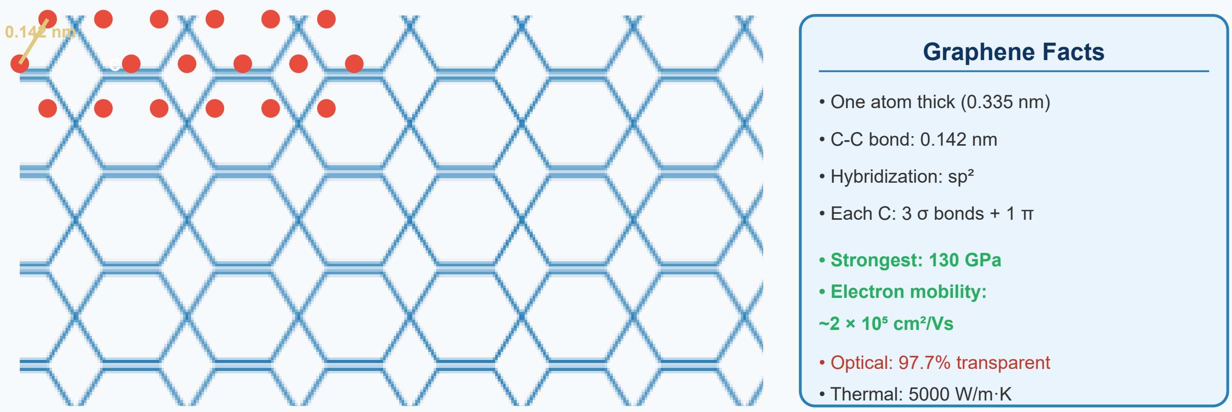



*Fig 15.1: Graphene honeycomb lattice — one atom thick layer of $sp^2$ hybridized carbon atoms arranged in hexagonal pattern. C-C bond length = 0.142 nm. Each carbon atom has one delocalized π electron contributing to extraordinary electronic properties.*

## 15.3 Properties of Graphene

| Property | Value | Comparison |
|---|---|---|
| Mechanical strength (Young's modulus) | ~1 TPa | ~5× stronger than steel |
| Tensile strength | ~130 GPa | Strongest material ever measured |
| Electron mobility | ~200,000 $cm^2/Vs$ | 100× better than silicon |
| Thermal conductivity | ~5000 W/m·K | 10× better than copper |
| Optical transparency | 97.7% | Nearly invisible |
| Density | ~0.77 $mg/m^2$ | Lightest material per unit area |
| Band gap | ~0 eV (zero gap semiconductor) | Massless Dirac fermion behavior |
| Surface area | ~2630 $m^2/g$ | Highest of any known material |

🏆 **GRAPHENE WORLD RECORDS**

• Thinnest material: 1 atom thick

• Strongest material: can bear ~42 N of force on a 1 $nm^2$ area

• Best electron conductor: electrons travel at 1/300 speed of light as massless Dirac fermions

• A graphene sheet 1 $m^2$ area weighs only ~0.77 mg

• One gram of graphene has a surface area larger than a tennis court (2630 $m^2$)

## 15.4 Applications of Graphene

| Application | Graphene Advantage Used | Status |
|---|---|---|
| Flexible electronics | Transparency + flexibility + conductivity | Commercial touchscreens emerging |

| | | |
|---|---|---|
| Ultra-fast transistors | 100× faster electron mobility than Si | Lab-scale, THz frequency achieved |
| Supercapacitors | 2630 $m^2/g$ surface area, high conductivity | Commercial products available |
| Composite materials | Adds extreme strength at low weight | Sports equipment, aerospace |
| Biomedical sensors | Sensitivity to single molecules | Research stage |
| Desalination membranes | Graphene oxide membranes filter ions precisely | Pilot projects |
| Photovoltaics | Transparent electrode replacing ITO | Research stage |
| Anti-corrosion coating | Impermeable to gases and liquids | Commercial applications |

## 15.5 CNT vs Graphene — Key Comparison

| Property | Carbon Nanotubes (CNT) | Graphene |
|---|---|---|
| Dimension | 1D (nanotube) | 2D (sheet) |
| Discovery | 1991 (Iijima) | 2004 (Geim & Novoselov) |
| Structure | Rolled graphene cylinder | Flat hexagonal $sp^2$ lattice |
| Electrical | Metallic or semiconducting (chirality-dependent) | Zero band gap semiconductor |
| Mechanical strength | ~63–150 GPa (tensile) | ~130 GPa (tensile) |
| Thermal conductivity | ~3500 W/m·K | ~5000 W/m·K |
| Surface area | ~1300 $m^2/g$ | ~2630 $m^2/g$ |
| Transparency | Not transparent | 97.7% transparent |
| Production cost | Moderate (CVD, arc discharge) | Higher (CVD on Cu) |

| | | |
|---|---|---|
| Best applications | Composites, drug delivery, sensors | Electronics, membranes, energy |

SECTION 16

# Solved Conceptual & Numerical Problems

40+ solved problems | Step-by-step solutions | Exam-oriented

## PART A — Semiconductor Problems

### Problem 1: Conductivity Calculation

**Q:** Calculate the conductivity of germanium (Ge) at 300 K given: $n_i = 2.4 \times 10^{13}$ cm$^{-3}$, $\mu_e = 3900$ cm$^2$/V·s, $\mu_h = 1900$ cm$^2$/V·s, $e = 1.6 \times 10^{-19}$ C

```
Given: ni = 2.4×10¹³ cm⁻³ | μₑ = 3900 cm²/Vs | μₕ = 1900 cm²/Vs | e = 1.6×10⁻¹⁹ C
```

**Formula:** $\sigma = n_i \cdot e \cdot (\mu_e + \mu_h)$

*(For intrinsic: n = p = ni, so both terms combine)*

$\sigma = 2.4\times10^{13} \times 1.6\times10^{-19} \times (3900 + 1900)$

$\sigma = 2.4\times10^{13} \times 1.6\times10^{-19} \times 5800$

$\sigma = 2.4 \times 1.6 \times 5800 \times 10^{13-19}$

$\sigma = 22272 \times 10^{-6}$

$\sigma = 2.227 \times 10^{-2}$ S/cm = **2.227 S/m**

✅ **Answer: $\sigma \approx 2.23$ S/m**

### Problem 2: Silicon Conductivity

**Q:** Find the conductivity of intrinsic silicon at 300 K. Given: $n_i = 1.5 \times 10^{10}$ cm$^{-3}$, $\mu_e = 1350$ cm$^2$/V·s, $\mu_h = 480$ cm$^2$/V·s

```
ni = 1.5×10¹⁰ cm⁻³ | μₑ = 1350 | μₕ = 480 cm²/Vs | e = 1.6×10⁻¹⁹ C
```

$\sigma = ni \cdot e \cdot (\mu_e + \mu_h) = 1.5\times10^{10} \times 1.6\times10^{-19} \times (1350 + 480)$

$\sigma = 1.5\times10^{10} \times 1.6\times10^{-19} \times 1830$

$\sigma = 1.5 \times 1.6 \times 1830 \times 10^{-9}$

$\sigma = 4392 \times 10^{-9}$ S/cm

$\sigma = 4.392 \times 10^{-6}$ S/cm = **$4.39 \times 10^{-4}$ S/m**

✅ **Answer: $\sigma \approx 4.39 \times 10^{-4}$ S/m (Si is much less conductive than Ge at room temp)**

### Problem 3: Resistivity of N-type Si

**Q:** Silicon is doped with phosphorus. Donor concentration Nd = $10^{16}$ cm$^{-3}$, $\mu_e$ = 1350 cm²/V·s. Find resistivity. (Assume ni << Nd, so n ≈ Nd)

```
Nd = 10¹⁶ cm⁻³ | μₑ = 1350 cm²/Vs | e = 1.6×10⁻¹⁹ C | n ≈ Nd (extrinsic dominates)
```

In n-type: $\sigma \approx n \cdot e \cdot \mu_e = Nd \cdot e \cdot \mu_e$

$\sigma = 10^{16} \times 1.6\times10^{-19} \times 1350$

$\sigma = 10^{16} \times 2.16\times10^{-16}$

$\sigma = 2.16$ S/cm = 216 S/m

Resistivity $\rho = 1/\sigma = 1/216$ = **$4.63 \times 10^{-3}$ Ω·m**

✅ **Answer: $\rho \approx 4.63 \times 10^{-3}$ Ω·m (much lower than intrinsic Si because of doping!)**

### Problem 4: Energy Gap from Conductivity

**Q:** The intrinsic carrier concentration of a semiconductor is $2.4\times10^{13}$/cm³ at 300 K. If Eg = 0.67 eV, calculate conductivity at 400 K (assume same mobilities).

```
T₁ = 300 K, ni₁ = 2.4×10¹³, Eg = 0.67 eV, k = 8.617×10⁻⁵ eV/K
```

Using: $ni \propto T^{(3/2)} \cdot \exp(-Eg/2kT)$

At 400K: $ni_2 = ni_1 \cdot (400/300)^{(3/2)} \cdot \exp[-(Eg/2k)(1/400 - 1/300)]^{-1}$

Ratio $= \exp[(Eg/2k)(1/T_1 - 1/T_2)]$

$= \exp[(0.67/2\times8.617\times10^{-5})(1/300 - 1/400)]$

$= \exp[3885 \times (3.33-2.5)\times10^{-3}]$

$= \exp[3885 \times 8.33\times10^{-4}] = \exp[3.24] \approx 25.5$

$ni_2 \approx 2.4\times10^{13} \times 25.5 \times (4/3)^{(3/2)} \approx 2.4\times10^{13} \times 25.5 \times 1.54 \approx 9.4\times10^{14}$ /cm³

✅ **ni increases significantly with temperature — confirming semiconductors get MORE conductive with heat.**

### Problem 5: Hole Concentration from Mass Action Law

**Q:** Silicon is doped with boron at $Na = 5\times10^{15}$ cm⁻³. Find the minority carrier (electron) concentration. Given $ni = 1.5\times10^{10}$ cm⁻³ at 300 K.

```
Na = 5×10¹⁵ cm⁻³, ni = 1.5×10¹⁰ cm⁻³
```

Mass Action Law: $n \cdot p = ni^2$

In p-type: $p \approx Na = 5\times10^{15}$ cm⁻³ (acceptor concentration dominates)

$n = ni^2 / p = (1.5\times10^{10})^2 / (5\times10^{15})$

$n = 2.25\times10^{20} / 5\times10^{15}$

$n = 4.5\times10^4$ cm⁻³

✅ **Answer: $n \approx 4.5\times10^4$ cm⁻³ (electrons are minority carriers in p-type — extremely rare!)**

## PART B — Superconductor Problems

### Problem 6: Critical Field Calculation

**Q:** For a superconductor with Tc = 9.25 K and Hc(0) = 160 kA/m, find the critical magnetic field at T = 6 K.

```
Tc = 9.25 K, Hc(0) = 160 kA/m, T = 6 K
```

> Formula: $H_c(T) = H_c(0)[1 - (T/T_c)^2]$
>
> $T/T_c = 6/9.25 = 0.6486$
>
> $(T/T_c)^2 = 0.421$
>
> $H_c(6) = 160 \times (1 - 0.421) = 160 \times 0.579$
>
> $H_c(6) = 92.6$ kA/m

✅ **Answer: Hc(6K) = 92.6 kA/m**

### Problem 7: Transition Temperature Identification

**Q:** A superconductor has Hc(0) = 200 kA/m. At T = 5 K, Hc(T) = 150 kA/m. Find its critical temperature Tc.

```
Hc(0) = 200, Hc(5K) = 150 kA/m, T = 5 K
```

> $H_c(T)/H_c(0) = 1 - (T/T_c)^2$
>
> $150/200 = 1 - (5/T_c)^2$
>
> $0.75 = 1 - (5/T_c)^2$
>
> $(5/T_c)^2 = 0.25$
>
> $5/T_c = 0.5$
>
> $T_c = 5/0.5 =$ **10 K**

✅ **Answer: Tc = 10 K**

### Problem 8: Isotope Effect

**Q:** Mercury isotope $^{200}Hg$ has Tc = 4.153 K. Using the isotope effect, estimate Tc for $^{198}Hg$. ($\alpha$ = 0.5 for Hg)

```
M₁ = 200, Tc₁ = 4.153 K, M₂ = 198, α = 0.5
```

Isotope Effect: $T_c \propto M^{(-\alpha)}$

$T_{c2}/T_{c1} = (M_1/M_2)^\alpha$

$T_{c2} = 4.153 \times (200/198)^{0.5}$

$T_{c2} = 4.153 \times (1.0101)^{0.5}$

$T_{c2} = 4.153 \times 1.00503$

$T_{c2} \approx$ **4.174 K**

✅ **Answer: $T_{c2} \approx 4.174$ K (lighter isotope → higher Tc, consistent with BCS theory)**

## Problem 9: Cooper Pair Energy

**Q:** Calculate the Cooper pair binding energy for Nb (Tc = 9.25 K). Given: $k_B = 1.38 \times 10^{-23}$ J/K

```
Tc = 9.25 K, kB = 1.38×10⁻²³ J/K, Cooper pair energy gap = 3.5 kBTc
(BCS result)
```

Energy gap ($2\Delta$) = $3.5\, k_B T_c$ (BCS weak coupling approximation)

$2\Delta = 3.5 \times 1.38\times10^{-23} \times 9.25$

$2\Delta = 3.5 \times 1.2765\times10^{-22}$

$2\Delta = 4.47\times10^{-22}$ J

$2\Delta = 4.47\times10^{-22} / 1.6\times10^{-19}$ eV = **2.79 meV**

✅ **Answer: Cooper pair energy gap ≈ 2.79 meV for Niobium**

# PART C — Supercapacitor Problems

### Problem 10: Energy Stored in Supercapacitor

**Q:** A supercapacitor has capacitance C = 3000 F and operating voltage V = 2.7 V. Calculate the energy stored.

```
C = 3000 F, V = 2.7 V
```

$E = \frac{1}{2} \cdot C \cdot V^2 = \frac{1}{2} \times 3000 \times (2.7)^2$

$E = 0.5 \times 3000 \times 7.29$

$E = 10{,}935$ J = **10.935 kJ**

In Wh: $E = 10935 / 3600 =$ **3.037 Wh**

✅ **Answer: E = 10.93 kJ = 3.04 Wh**

### Problem 11: Capacitance from EDL Formula

**Q:** An EDLC electrode has surface area $A = 1000\ m^2$, double layer thickness d = 0.5 nm, $\varepsilon_r = 10$. Calculate capacitance. $\varepsilon_0 = 8.85\times10^{-12}$ F/m.

```
A = 1000 m², d = 0.5×10⁻⁹ m, εᵣ = 10, ε₀ = 8.85×10⁻¹² F/m
```

$C = \varepsilon_0 \cdot \varepsilon_r \cdot A / d$

$C = 8.85\times10^{-12} \times 10 \times 1000 / (0.5\times10^{-9})$

$C = 8.85\times10^{-8} / 0.5\times10^{-9}$

$C = 8.85\times10^{-8} / 5\times10^{-10}$

$C = 177$ F

✅ **Answer: C = 177 F (the massive surface area + tiny gap gives enormous capacitance!)**

### Problem 12: Specific Capacitance

**Q:** An EDLC electrode made of activated carbon has mass m = 5g and capacitance C = 250 F. Find specific capacitance.

Specific Capacitance = C / m = 250 F / 5 g = **50 F/g**

This is typical for commercial activated carbon EDLCs (range: 100–300 F/g for high-surface-area carbon).

✅ **Answer: Specific Capacitance = 50 F/g**

### Problem 13: Power Density

**Q:** A supercapacitor with V = 2.5 V and ESR = 0.5 mΩ. Find its peak power density for a 10g device.

```
V = 2.5 V, ESR = 0.5×10⁻³ Ω, m = 0.01 kg
```

$P_{max} = V^2 / (4 \times ESR) = (2.5)^2 / (4 \times 0.5\times10^{-3})$

$= 6.25 / (2\times10^{-3}) = 3125$ W

Specific Power = P / m = 3125 / 0.01 = **312,500 W/kg = 312.5 kW/kg**

✅ **Answer: Power density ≈ 312.5 kW/kg (compare: Li-ion battery ≈ 0.15–0.5 kW/kg!)**

## PART D — Nanomaterial Conceptual Problems

### Problem 14: Surface-to-Volume Ratio

**Q:** Compare the surface-to-volume ratio of a cube with side 1 cm vs a nanocube with side 10 nm.

For a cube of side a: $S/V = 6a^2/a^3 = 6/a$

Bulk cube (a = 1 cm = 0.01 m): $S/V = 6/0.01 = 600\ m^{-1}$

Nanocube (a = 10 nm = $10\times10^{-9}$ m): $S/V = 6/(10\times10^{-9}) = 6\times10^8\ m^{-1}$

Ratio = $6\times10^8$ / 600 = **$10^6$ times larger**

✅ **The nanocube has a surface-to-volume ratio ONE MILLION times larger than the bulk cube — this is why nano-catalysts are so effective!**

### Problem 15: Quantum Confinement Energy

**Q:** Estimate the lowest energy of an electron confined in a 1D box of size L = 2 nm (a quantum dot). $m^* = 0.07 \times m_0$ (GaAs). ($h = 6.626\times10^{-34}$ J·s, $m_0 = 9.11\times10^{-31}$ kg)

```
L = 2×10⁻⁹ m, m* = 0.07×9.11×10⁻³¹ = 6.38×10⁻³² kg
```

> Particle in a box: $E_1 = h^2 / (8m^*L^2)$
>
> $E_1 = (6.626\times10^{-34})^2 / (8 \times 6.38\times10^{-32} \times (2\times10^{-9})^2)$
>
> $= 4.39\times10^{-67} / (8 \times 6.38\times10^{-32} \times 4\times10^{-18})$
>
> $= 4.39\times10^{-67} / (2.04\times10^{-48})$
>
> $= 2.15\times10^{-19}$ J $= 2.15\times10^{-19} / 1.6\times10^{-19}$ eV = **1.34 eV**

✅ **The confined electron has energy 1.34 eV — much higher than bulk GaAs band gap of 1.42 eV. Quantum dots can be tuned to emit specific colors by changing size!**

### Problem 16: CNT Circumference and Diameter

**Q:** For a SWCNT with chiral vector (10,10) (armchair), find its diameter. C-C bond length = 0.142 nm.

> For chiral vector (n,m): $|C| = a\sqrt{(n^2 + nm + m^2)}$ where a = 0.246 nm (graphene lattice constant)
>
> For (10,10): $|C| = 0.246 \times \sqrt{(100 + 100 + 100)} = 0.246 \times \sqrt{300} = 0.246 \times 17.32 = 4.26$ nm
>
> Diameter $d = |C|/\pi = 4.26/\pi$ = **1.36 nm**
>
> Since n = m (armchair configuration) → this CNT is METALLIC

✅ **Answer: d ≈ 1.36 nm; (10,10) armchair CNT = metallic conductor**

### Problem 17: Hall Coefficient

**Q:** The Hall coefficient of a semiconductor is RH = $3.125 \times 10^{-4}$ $m^3/C$. Identify the carrier type and find carrier concentration.

Hall coefficient RH = 1/(n·e) for n-type OR −1/(p·e) for p-type

If RH is positive ($+3.125 \times 10^{-4}$) → material is P-TYPE (holes are majority)

$p = 1/(RH \times e) = 1/(3.125 \times 10^{-4} \times 1.6 \times 10^{-19})$

$p = 1/(5 \times 10^{-23})$ = **$2 \times 10^{22}$ $m^{-3}$ = $2 \times 10^{16}$ $cm^{-3}$**

✅ **P-type semiconductor; hole concentration p = $2 \times 10^{16}$ $cm^{-3}$**

### Problem 18: Fullerene Molecular Weight

**Q:** Calculate the molecular weight of $C_{60}$ fullerene. (Atomic mass of C = 12 g/mol)

$C_{60}$ has 60 carbon atoms

Molecular weight = 60 × 12 = **720 g/mol**

This is confirmed by mass spectrometry — m/z = 720 peak identifies $C_{60}$ uniquely.

✅ **Answer: MW of $C_{60}$ = 720 g/mol**

### Problem 19: Energy Density Comparison

**Q:** A supercapacitor (C = 500F, V = 2.7V) and a Li-ion battery (V = 3.6V, capacity = 2 Ah) are compared. Which has higher energy density? (Assume both weigh 100g)

Supercapacitor: $E = ½CV^2 = 0.5 \times 500 \times 2.7^2 = 0.5 \times 500 \times 7.29 = 1822.5$ J

Energy density = 1822.5 J / 0.1 kg = 18,225 J/kg = 18.2 kJ/kg = **5.06 Wh/kg**

Li-ion battery: E = V × Q = 3.6 V × 2 Ah = 7.2 Wh = 25,920 J

Energy density = 25920 / 0.1 = 259,200 J/kg = **72 Wh/kg**

Battery energy density = 72 Wh/kg; Supercapacitor = 5.06 Wh/kg

✅ **Battery has ~14× higher energy density. But supercapacitor has ~1000× higher power density! They serve different purposes.**

### Problem 20: Band Gap Energy Conversion

**Q:** A semiconductor has band gap Eg = 1.42 eV (GaAs). (a) Convert to Joules. (b) Find minimum frequency of photon that can excite an electron across the gap.

> (a) $E_g = 1.42 \text{ eV} \times 1.6\times10^{-19}$ J/eV = $\mathbf{2.272\times10^{-19}}$ **J**
>
> (b) $E = hf \rightarrow f = E/h = 2.272\times10^{-19} / 6.626\times10^{-34}$
>
> $f = 3.43\times10^{14}$ Hz (near-infrared / visible light regime)
>
> Wavelength $\lambda = c/f = 3\times10^{8}/3.43\times10^{14}$ = **875 nm** (infrared)

✅ **GaAs absorbs photons with λ < 875 nm — this is why it's used in IR photodetectors and laser diodes!**

### Problems 21–25: Quick Calculations

**21.** If $\sigma = 50$ S/m for a semiconductor, find resistivity. → $\rho = 1/50$ = **0.02 Ω·m**

**22.** A superconductor has Tc = 7.2 K. At what temperature is $H_c = 0.75 \times H_c(0)$? → Using $H_c(T)/H_c(0) = 1-(T/T_c)^2$: $T = T_c\times\sqrt{0.25} = 7.2\times0.5$ = **3.6 K**

**23.** Graphene electron mobility is $2\times10^{5}$ $cm^2/Vs$. If carrier density $n = 10^{12}$ $cm^{-2}$, find sheet conductivity. $\sigma = n\cdot e\cdot\mu = 10^{12}\times10^{4}\times1.6\times10^{-19}\times2\times10^{5}\times10^{4}$ = **320 S**

**24.** What is the percentage of surface atoms in a 10 nm Au nanoparticle (atom radius ~0.14 nm)? N_surface/N_total ≈ $(d+2r)^3/d^3-1$ approach; for 10nm ~40% atoms are on surface.

**25.** How many C atoms in a 1 g sample of $C_{60}$? → MW = 720, N_Avogadro = $6.022\times10^{23}$; Moles = 1/720; C atoms = $(1/720)\times6.022\times10^{23}\times60$ = $\mathbf{5.02\times10^{22}}$

### Problems 26–30: Conceptual Application

**26.** Why does silicon (Eg=1.12 eV) perform better than Ge (Eg=0.67 eV) for high-temperature applications?
*Ans: Larger band gap → fewer thermally generated carriers at high T → less thermal noise*

*and leakage → more stable operation. Si remains semiconductor up to ~175°C; Ge only ~75°C.*

**27.** Why are SWCNTs sometimes metallic and sometimes semiconducting?
*Ans: Depends on chirality (rolling angle). Armchair (n=m): metallic. Zigzag/chiral: can be either. ~1/3 are metallic, 2/3 semiconducting.*

**28.** Why does the Meissner effect distinguish superconductors from "perfect conductors"?
*Ans: A perfect conductor would trap existing magnetic flux; a superconductor actively expels B regardless of whether field was applied before or after cooling. Meissner effect shows superconducting state is a distinct thermodynamic state.*

**29.** Why is graphene called a "zero-gap semiconductor"?
*Ans: Conduction and valence bands touch at exactly 2 points (Dirac points) in k-space but with zero density of states at Fermi level. Not metallic (no overlapping bands) nor true semiconductor (no gap). Electrons behave as massless Dirac fermions.*

**30.** Why do supercapacitors have better cycle life than batteries?
*Ans: No chemical reactions occur — charge storage is purely electrostatic (EDLC) or surface faradaic. No bulk electrode degradation, volume changes, or irreversible chemical transformations. Can cycle 1 million+ times vs 500–2000 for Li-ion.*

## Problems 31–40: Additional Solved Problems

**31.** Find mobility of electrons if $\sigma = 100$ S/m, $n = 8\times10^{20}$ m$^{-3}$, $e = 1.6\times10^{-19}$ C.
$\mu_e = \sigma/(n\cdot e) = 100/(8\times10^{20}\times1.6\times10^{-19}) = 100/128 =$ **0.781 m²/Vs = 7810 cm²/Vs**

**32.** A supercapacitor is charged to 2.5V with C = 100F. How much energy is released if discharged to 1V?
$E = \frac{1}{2}C(V_1^2-V_2^2) = 0.5\times100\times(6.25-1) = 0.5\times100\times5.25 =$ **262.5 J**

**33.** Josephson junction: What is the frequency of AC Josephson oscillation when V = 1 mV is applied?
$f = 2eV/h = 2\times1.6\times10^{-19}\times10^{-3}/6.626\times10^{-34} = 3.2\times10^{-22}/6.626\times10^{-34} =$ **$4.83\times10^{11}$ Hz = 483 GHz**

**34.** How many hexagons and pentagons are in $C_{70}$?
$C_{70}$: Euler theorem: 12 pentagons (always) + 25 hexagons → Total = 12×5/2 + 25×6/2 + crossings... For $C_{70}$: **12 pentagons, 25 hexagons**

**35.** An n-type Si sample has ND = $10^{17}$/cm³. Find resistivity. $\mu_e$ = 700 cm²/Vs (reduced due to impurity scattering).
$\sigma = ND\times e\times\mu_e = 10^{17}\times1.6\times10^{-19}\times700 = 11.2$ S/cm; $\rho = 1/11.2 =$ **0.089 Ω·cm**

**36.** What minimum surface area of activated carbon is needed to make a 1000F EDLC electrode? ($d=0.8nm$, $\varepsilon_r=12$)

$C = \varepsilon_0\varepsilon_r A/d \rightarrow A = Cd/(\varepsilon_0\varepsilon_r) = 1000\times0.8\times10^{-9}/(8.85\times10^{-12}\times12) = 8\times10^{-7}/1.062\times10^{-10} =$ **7534 $m^2$**

**37.** London penetration depth for Pb at 0K is $\lambda(0) = 37$ nm. Find $\lambda$ at $T = 6.5$ K ($T_c = 7.19$ K).

$\lambda(T) = \lambda(0)/[1-(T/Tc)^4]$^0.5 = $37/[1-(6.5/7.19)^4]$^0.5 = $37/[1-0.66]$^0.5 = $37/0.583 =$ **63.5 nm**

**38.** If a CNT has thermal conductivity 3500 W/m·K and cross-section $A = \pi(1nm)^2 = 3.14\times10^{-18}\ m^2$, find thermal conductance for 1 μm length.

$G = kA/L = 3500\times3.14\times10^{-18}/10^{-6} =$ **$1.099\times10^{-8}$ W/K = 11 nW/K**

**39.** A graphene FET has channel mobility $\mu = 10{,}000\ cm^2/Vs$, carrier density $n = 10^{12}\ cm^{-2}$, channel 1μm × 1μm. Find channel resistance.

$\sigma = n\cdot e\cdot\mu$ (sheet) $= 10^{12}\times1.6\times10^{-19}\times10^4\times10^4 = 16$ S (sheet conductance).

$R = 1/(\sigma \times W/L) = 1/16 =$ **62.5 mΩ**

**40.** Calculate the de Broglie wavelength of an electron in a quantum dot at room temperature (KE = 3/2 kBT).

$KE = 1.5 \times 1.38\times10^{-23} \times 300 = 6.21\times10^{-21}$ J; $\lambda = h/\sqrt{(2m\cdot KE)} = 6.626\times10^{-34}/\sqrt{(2\times9.11\times10^{-31}\times6.21\times10^{-21})} = 6.626\times10^{-34}/\sqrt{(1.132\times10^{-50})} = 6.626\times10^{-34}/3.36\times10^{-25} =$ **1.97 nm** (Consistent with quantum confinement at ~2nm scale!)

SECTION 17

# Questions & Answers (2M & 5M)

Short answers | Detailed answers | Exam-pattern questions

## PART A — 2 Mark Questions (Short Answers)

### Q1. Define a semiconductor. 2M

**Ans:** A semiconductor is a material with electrical conductivity between that of a conductor and an insulator. It has a band gap (Eg) typically between 0.1 to 3.0 eV and its conductivity increases with temperature. Examples: Silicon (Eg = 1.12 eV), Germanium (Eg = 0.67 eV).

### Q2. What is doping? 2M

**Ans:** Doping is the intentional introduction of specific impurity atoms (dopants) into a pure semiconductor in very small quantities (parts per million) to modify its electrical conductivity. Group V atoms create n-type (electrons), Group III atoms create p-type (holes).

### Q3. State the Meissner effect. 2M

**Ans:** The Meissner effect is the complete expulsion of magnetic flux from the interior of a superconductor when it is cooled below its critical temperature (Tc). A superconductor exhibits perfect diamagnetism (B = 0 inside), which is distinct from merely having zero resistance.

### Q4. What is the critical temperature (Tc) of a superconductor? 2M

**Ans:** Critical temperature (Tc) is the characteristic temperature below which a material transitions from normal conducting behavior to superconducting state (zero resistance). Above Tc, the material has normal resistance. Examples: Hg: 4.15 K; YBCO: 92 K; Mercury cuprate: 138 K (highest known).

### Q5. What are Cooper pairs? 2M

**Ans:** Cooper pairs are bound pairs of electrons in a superconductor, formed through an attractive interaction mediated by lattice vibrations (phonons). The two electrons have opposite spins and momenta. These pairs act as bosons and form a coherent quantum condensate that flows without resistance, according to BCS theory.

### Q6. Distinguish Type I and Type II superconductors. 2M

**Ans:** Type I has a single critical field (Hc) — complete Meissner effect below it, normal above it. Pure metals (Hg, Pb). Type II has two critical fields (Hc1 and Hc2) — complete Meissner below Hc1, mixed/vortex state between Hc1 and Hc2, normal above Hc2. Alloys/ceramics (NbTi, YBCO). Type II are used in practical applications.

### Q7. What is an EDLC? 2M

**Ans:** An Electric Double Layer Capacitor (EDLC) is a type of supercapacitor that stores energy electrostatically by adsorption of electrolyte ions onto high-surface-area carbon electrodes, forming an Electric Double Layer at the electrode-electrolyte interface. No chemical reactions occur, enabling very long cycle life (>1 million cycles).

### Q8. What are nanomaterials? Give their size range. 2M

**Ans:** Nanomaterials are materials with at least one dimension in the range of 1–100 nanometers (1 nm = $10^{-9}$ m). At this scale, quantum mechanical effects become dominant and materials exhibit unique properties different from bulk. Examples: Gold nanoparticles (0D), CNT (1D), Graphene (2D).

### Q9. What is Buckminsterfullerene? 2M

**Ans:** Buckminsterfullerene ($C_{60}$) is a spherical carbon allotrope consisting of 60 carbon atoms arranged in 20 hexagons and 12 pentagons, resembling a soccer ball. Named after architect R. Buckminster Fuller. Diameter ~0.7 nm. Discovered in 1985 by Kroto, Curl, and Smalley (Nobel Prize 1996).

### Q10. What is the significance of chirality in CNTs? 2M

**Ans:** Chirality (rolling angle) of a SWCNT determines its electronic properties. Described by chiral vector (n,m): armchair CNTs (n=m) are metallic; zigzag/chiral CNTs can be metallic or semiconducting based on whether (n-m) is divisible by 3. About 1/3 of SWCNTs are metallic and 2/3 are semiconducting.

### Q11. Why is graphene called the "thinnest material"? 2M

**Ans:** Graphene consists of a single layer of carbon atoms arranged in a 2D hexagonal lattice. Its thickness is exactly one atom (0.335 nm = 3.35 Å), making it literally the thinnest possible material — you cannot make a material thinner than one atom. First isolated by mechanical exfoliation using Scotch tape in 2004.

### Q12. What is the mass action law for semiconductors? 2M

**Ans:** The Mass Action Law states that the product of electron concentration (n) and hole concentration (p) in a semiconductor at thermal equilibrium equals the square of intrinsic carrier concentration (ni): $\mathbf{n \times p = ni^2}$. This holds regardless of doping type and is useful to find minority carrier concentrations.

## PART B — 5 Mark Questions (Detailed Answers)

### Q13. Explain the band theory of solids and classify materials based on band gap. 5M

**Band Theory of Solids:**

When atoms come together to form a solid, atomic energy levels split into broad energy bands due to Pauli exclusion principle and inter-atomic interactions. The key bands are:

**1. Valence Band (VB):** Highest completely filled energy band at absolute zero. Electrons here are bound to atoms and don't contribute to conduction.

**2. Conduction Band (CB):** The next higher energy band, empty or partially filled. Free electrons here conduct electricity.

**3. Forbidden Gap (Eg):** Energy gap between top of VB and bottom of CB. No allowed energy states exist here.

**Classification:**

• **Conductors:** Overlapping VB and CB (Eg ≈ 0). Electrons freely available. e.g., Cu, Al, Ag.

• **Semiconductors:** Small forbidden gap (Eg = 0.1–3 eV). Some electrons cross the gap at room temperature. e.g., Si (1.12 eV), Ge (0.67 eV), GaAs (1.42 eV).

• **Insulators:** Large forbidden gap (Eg > 5 eV). Electrons cannot cross at room temperature. e.g., Diamond (5.5 eV), glass (~10 eV).

**Key Conclusion:** The band gap width determines whether a material conducts electricity, and in semiconductors, conductivity can be tuned by temperature, light, and doping.

## Q14. Describe n-type and p-type semiconductors with neat diagrams. 5M

**Extrinsic Semiconductors — formed by doping pure semiconductors:**

**N-type Semiconductor:**

Formed by doping with Group V elements (P, As, Sb — pentavalent).
The dopant atom has 5 valence electrons — 4 form covalent bonds with Si, the 5th becomes a FREE electron with very little energy needed.
These atoms are called DONOR atoms (donate electrons to CB).
Majority carriers: Electrons (n >> p). Minority carriers: Holes.
Fermi level shifts toward conduction band.
Conductivity: $\sigma \approx N_d \times e \times \mu_e$ (where Nd = donor concentration)

**P-type Semiconductor:**

Formed by doping with Group III elements (B, Ga, In — trivalent).
The dopant atom has 3 valence electrons — forms only 3 covalent bonds, creating an

empty position (HOLE).
These atoms are called ACCEPTOR atoms (accept electrons from VB, creating holes).
Majority carriers: Holes (p >> n). Minority carriers: Electrons.
Fermi level shifts toward valence band.
Conductivity: $\sigma \approx N_a \times e \times \mu_h$ (where Na = acceptor concentration)

**Key Formula:** $n \times p = n_i^2$ (Mass Action Law holds for both types)

**Applications:** p-n junction (diode), transistors, solar cells, LEDs — all require controlled doping.

## Q15. Write a detailed note on superconductors — properties and applications. 5M

**Superconductors — Zero Resistance Materials:**

**Definition:** Materials showing zero electrical resistance and complete expulsion of magnetic field below critical temperature Tc.

**Properties:**

1. **Zero Resistance:** R = 0 below Tc. Current flows indefinitely without energy loss. Demonstrated by persistent current experiments lasting years.

2. **Meissner Effect:** Complete expulsion of magnetic flux (B = 0) from interior. Superconductor is a perfect diamagnet. Causes magnetic levitation.

3. **Critical Temperature (Tc):** Transition temperature from normal to superconducting state. Mercury: 4.15 K; YBCO: 92 K; Record: 138 K.

4. **Critical Magnetic Field (Hc):** Maximum field the superconductor can tolerate. $H_c(T) = H_c(0)[1-(T/T_c)^2]$.

5. **Critical Current (Jc):** Maximum current density maintaining superconductivity.

6. **Josephson Effect:** Quantum tunneling of Cooper pairs through thin insulating barrier — used in SQUID sensors.

**BCS Theory:** Superconductivity explained by Cooper pair formation — electrons pair via phonon-mediated interaction, forming bosonic condensate with no scattering.

**Applications:**

• MRI machines — NbTi coils create strong stable magnetic fields

• Maglev trains — Meissner levitation eliminates friction (Japan SCMaglev: 603 km/h)

- LHC particle accelerator — 1232 superconducting dipole magnets
- SQUID sensors — detect magnetic fields of $10^{-18}$ Tesla (brain activity!)
- Superconducting power cables — zero loss electricity transmission
- Quantum computers — Josephson junction qubits (IBM, Google)

## Q16. Explain the working mechanism of supercapacitors. Classify them. 5M

**Supercapacitors — High-Energy Capacitors:**

**Classification:**

1. **EDLC (Electric Double Layer Capacitors):** Non-Faradaic, purely electrostatic charge storage at electrode surface. Carbon electrodes. 1M+ cycle life.

2. **Pseudocapacitors:** Faradaic (fast surface redox reactions). Metal oxides ($RuO_2$, $MnO_2$) or conducting polymers. Higher energy density than EDLC.

3. **Hybrid Supercapacitors:** Combine EDLC and battery/pseudocapacitor electrodes. Best of both: high energy + high power.

**Working Mechanism of EDLC:**

**Step 1 — Structure:** Two porous carbon electrodes, electrolyte, separator between them, current collectors.

**Step 2 — Charging:** Applied voltage drives cations toward negative electrode and anions toward positive electrode. They adsorb on electrode surfaces forming the Electric Double Layer (EDL) — two layers: (i) Stern layer — compact adsorbed ions; (ii) Diffuse layer — loosely held ions.

**Step 3 — Energy storage:** Energy stored = $\frac{1}{2}CV^2$. The capacitance is enormous because: (a) electrode area A is huge (1000–3000 $m^2/g$); (b) EDL thickness d is extremely small (0.5–1 nm). $C = \varepsilon_0\varepsilon_r A/d \rightarrow$ Very large C.

**Step 4 — Discharging:** Ions desorb, return to electrolyte, electrons flow through external circuit. No chemical change — fully reversible.

**Applications:** EVs (regenerative braking), UPS, wind/solar energy smoothing, camera flash units, start-stop engines, memory backup, portable electronics.

## Q17. Compare properties and applications of Fullerenes, CNTs, and Graphene. 5M

**Comparison of Carbon Nanomaterials:**

**Fullerenes ($C_{60}$):**
Structure: Spherical cage (0D) — 60 C atoms, 20 hexagons + 12 pentagons. Diameter 0.7 nm.
Discovery: Kroto, Curl, Smalley (1985; Nobel 1996)
Properties: Semiconductor (~1.9 eV gap), excellent electron acceptor, can be made superconducting ($K_3C_{60}$, Tc=18K), soluble in organic solvents, can encapsulate atoms/drugs.
Applications: Drug delivery, organic solar cells (PCBM), antioxidants, lubricants, HIV drug development.

**Carbon Nanotubes (CNTs):**
Structure: Rolled graphene cylinder (1D) — SWCNT (0.4–2nm dia) or MWCNT (2–100nm).
Discovery: Iijima (1991, MWCNT; 1993, SWCNT)
Properties: Tensile strength ~63–150 GPa (100× steel), Young's modulus ~1 TPa, thermal conductivity ~3500 W/mK, electrical: metallic or semiconducting (chirality-dependent), electron mobility ~100,000 $cm^2/Vs$.
Applications: Composite materials, transistors, drug delivery, sensors, supercapacitor electrodes, field emission displays, thermal interface materials.

**Graphene:**
Structure: Single atomic layer 2D hexagonal lattice of $sp^2$ carbon. Thickness 0.335 nm.
Discovery: Geim and Novoselov (2004; Nobel Physics 2010)
Properties: Strongest material (130 GPa tensile), highest electron mobility (~200,000 $cm^2/Vs$), best thermal conductor (~5000 W/mK), 97.7% optically transparent, zero band gap semiconductor, surface area ~2630 $m^2/g$.
Applications: Flexible electronics, ultra-fast transistors, supercapacitors, composite materials, biomedical sensors, water purification membranes, anti-corrosion coatings, touchscreens.

## Q18. What are nanomaterials? Classify them and explain their unique properties. 5M

**Nanomaterials:** Materials with at least one dimension in the 1–100 nm range (1 nm = $10^{-9}$ m). At nanoscale, quantum effects dominate and properties differ dramatically from bulk.

**Classification by Dimensionality:**

• **0D (Zero-dimensional):** All 3 dimensions at nanoscale. Nanoparticles, quantum dots. Examples: Au nanoparticles, $C_{60}$, semiconductor quantum dots.

• **1D (One-dimensional):** One dimension extends beyond nanoscale. Nanowires, nanorods, nanotubes. Examples: CNT, ZnO nanowires, silver nanowires.

• **2D (Two-dimensional):** Two dimensions extend; thickness is nano. Nanosheets, thin films. Examples: Graphene, $MoS_2$, h-BN, thin films.

• **3D (Three-dimensional):** All dimensions > 100nm but nanograined structure. Nanocrystalline metals, nanocomposites.

**Unique Properties of Nanomaterials:**

1. **Quantum Size Effect:** Electron energy levels become quantized (discrete) when confined to nanoscale dimensions. This changes optical, electronic, and magnetic properties. Gold nanoparticles appear red (not gold) due to surface plasmon resonance.

2. **High Surface-to-Volume Ratio:** A 10 nm nanoparticle has ~40% of atoms on the surface vs. 0.04% for 1 μm particle. Makes nanomaterials extremely reactive — ideal catalysts.

3. **Size-dependent Properties:** Melting point decreases (Au melts at ~300°C at 2nm vs 1064°C in bulk). Hardness can increase dramatically (Hall-Petch relationship).

4. **Optical Properties:** Quantum dots emit different colors based on size — same material, different color! Tunable from UV to IR by size control.

5. **Magnetic Properties:** Ferromagnetic materials become superparamagnetic below ~20 nm — no permanent magnetism but instant magnetization by external field.

SECTION 18

# ⚡ Quick Revision Sheet

All key formulas | Important concepts | Exam shortcuts | Last-minute review

## SEMICONDUCTORS — Key Summary

| | |
|---|---|
| **Conductivity** | $\sigma = n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h$ |
| **Mass Action Law** | $n \cdot p = n_i^2$ |
| **Resistivity** | $\rho = 1/\sigma$ |
| **Band gap Si** | $E_g = 1.12$ eV |
| **Band gap Ge** | $E_g = 0.67$ eV |
| **N-type dopant** | Group V: P, As, Sb → extra $e^-$ |
| **P-type dopant** | Group III: B, Ga, In → creates $h^+$ |
| **Temp effect** | Conductivity ↑ with T (opposite to metals) |

$\sigma = \sigma_0 \cdot \exp(-E_g / 2kT)$ — exponential T-dependence

## SUPERCONDUCTORS — Key Summary

| | |
|---|---|
| **Critical field** | $H_c(T) = H_c(0)[1 - (T/T_c)^2]$ |
| **Meissner effect** | $B = 0$ inside superconductor |
| **Isotope effect** | $T_c \propto M^{-0.5}$ |
| **Energy gap (BCS)** | $2\Delta = 3.5\ k_B T_c$ |
| **AC Josephson** | $f = 2eV/h$ |

| | |
|---|---|
| **YBCO Tc** | 92 K (liquid $N_2$ cooling) |
| **Type I vs II** | Single Hc vs Two critical fields (Hc1, Hc2) |

```
3 Critical Parameters: Tc (temperature) + Hc (field) + Jc (current) = "Critical Triangle"
```

## 🔋 SUPERCAPACITORS — Key Summary

| | |
|---|---|
| **Capacitance** | $C = \varepsilon_0 \cdot \varepsilon_r \cdot A/d$ |
| **Energy stored** | $E = \frac{1}{2} \cdot C \cdot V^2$ |
| **Peak power** | $P = V^2/(4 \cdot ESR)$ |
| **EDLC** | Non-Faradaic, electrostatic, 1M+ cycles |
| **Pseudocapacitor** | Faradaic, surface redox, higher energy |
| **Hybrid** | EDLC + battery electrode — best of both |
| **Materials** | Activated carbon, $RuO_2$, $MnO_2$, graphene |

```
Key advantage: Very high power density (up to 100 kW/kg) vs battery (0.1-0.5 kW/kg)
```

## 🔬 NANOMATERIALS — Key Summary

| | |
|---|---|
| **Size range** | 1–100 nm (1 nm = $10^{-9}$ m) |
| **S/V ratio** | For cube: $S/V = 6/a$ (increases as size decreases) |
| **0D** | Nanoparticles, quantum dots, $C_{60}$ |
| **1D** | CNT, nanowires, nanorods |
| **2D** | Graphene, $MoS_2$, thin films |
| **Quantum dot energy** | $E_1 = h^2/(8m^*L^2)$ |
| **C60** | 60 C atoms, 20 hexagons, 12 pentagons, 0.7 nm dia |

| | |
|---|---|
| **CNT chirality** | n=m → metallic; others may be semiconducting |
| **Graphene** | 1 atom thick, 130 GPa strength, 2630 $m^2/g$, 97.7% transparent |

## 📊 COMPARISON TABLES — Quick Reference

### Semiconductor vs Superconductor:

| Feature | Semiconductor | Superconductor |
|---|---|---|
| Resistance | Finite (variable) | Zero (below Tc) |
| Temp effect on R | R decreases with T↑ | R → 0 abruptly at Tc |
| Magnetic field | Not affected | Expelled (Meissner) |
| Applications | Chips, solar cells, LEDs | MRI, maglev, SQUID |

### CNT vs Graphene:

| Feature | CNT | Graphene |
|---|---|---|
| Dimension | 1D cylinder | 2D sheet |
| Strength | ~63–150 GPa | ~130 GPa |
| Thermal (W/mK) | ~3500 | ~5000 |
| Transparency | Opaque | 97.7% transparent |
| Electrical | Metallic or SC (chirality) | Zero-gap semimetal |

## 🏆 IMPORTANT SCIENTISTS & DISCOVERIES

| Discovery | Scientist(s) | Year | Award |
|---|---|---|---|
| Superconductivity (Hg) | Kamerlingh Onnes | 1911 | Nobel Physics 1913 |
| Meissner Effect | Meissner & Ochsenfeld | 1933 | — |
| BCS Theory | Bardeen, Cooper, Schrieffer | 1957 | Nobel Physics 1972 |

| | | | |
|---|---|---|---|
| HTS (La-Ba-Cu-O) | Bednorz & Müller | 1986 | Nobel Physics 1987 |
| Fullerenes ($C_{60}$) | Kroto, Curl, Smalley | 1985 | Nobel Chemistry 1996 |
| Carbon Nanotubes | Sumio Iijima | 1991 | — |
| Graphene isolation | Geim & Novoselov | 2004 | Nobel Physics 2010 |

## ⚡ MOST IMPORTANT FORMULAS — EXAM CHEAT SHEET

```
σ = n·e·μₑ + p·e·μₕ    [Semiconductor conductivity]
```

```
n·p = ni²    [Mass action law]
```

```
Hc(T) = Hc(0)[1 − (T/Tc)²]    [Superconductor critical field]
```

```
B = 0 (inside superconductor)    [Meissner Effect]
```

```
E = ½·C·V²    [Supercapacitor energy]
```

```
C = ε₀·εᵣ·A/d    [EDL Capacitance]
```

```
E₁ = h²/(8m*L²)    [Quantum confinement energy]
```

```
S/V = 6/a (cube)    [Surface-to-volume ratio]
```

```
2Δ = 3.5kBTc    [BCS Cooper pair energy gap]
```

```
f = 2eV/h    [Josephson AC frequency]
```

## 🎯 ONE-LINE DEFINITIONS — EXAM ESSENTIALS

• **Semiconductor:** Material with Eg = 0.1–3 eV; conductivity increases with temperature

• **Doping:** Adding controlled impurities to change semiconductor conductivity

• **p-n Junction:** Interface of p-type and n-type semiconductor; basis of all semiconductor devices

• **Superconductor:** Zero resistance + Meissner effect below Tc

• **Cooper Pairs:** Electron pairs (via phonons) that carry current without scattering in superconductors

• **Meissner Effect:** Complete expulsion of magnetic flux (B=0) inside superconductor below Tc

• **EDLC:** Supercapacitor storing charge in Electric Double Layer (non-Faradaic)

• **Nanomaterial:** Material with ≥1 dimension in 1–100 nm range

• **Quantum confinement:** Quantization of energy levels when electrons confined to nanoscale

• **Fullerene:** Spherical carbon cage ($C_{60}$: 60 C atoms, 20 hexagons + 12 pentagons)

• **CNT:** Rolled graphene cylinder; SWCNT (0.4–2nm) or MWCNT (2–100nm)

• **Graphene:** Single-atom-thick 2D hexagonal carbon lattice; strongest + most conductive material